SELF ASSESSMENT 1-1 (Pages 10 – 11)

1. $\Delta T1\ ^\circ C \equiv \Delta T1\ K$

$\therefore \Delta T15\ ^\circ C \equiv \Delta T15\ K$

$\therefore$ temperature rise of 15 °C in kelvin is **15 K**

2. $P_1 + \rho_w g h_1 + \rho_o g h_2 - \rho_m g h_3 = P_{atm}$

Rearranging fo P_1

$P_1 = P_{atm} - \rho_w g h_1 - \rho_o g h_2 + \rho_m g h_3$

The gage pressure of air in the tank is given by:

$P_1 - P_{atm} = g[\rho_m h_3 - \rho_w h_1 - \rho_o h_2]$

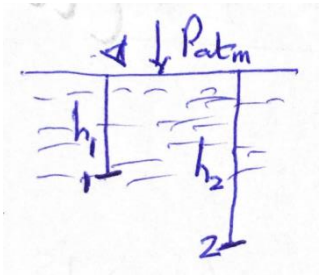
Substituting we have,

$$P_1 - P_{atm} = 9.81[13600 \times 0.46 - 1000 \times 0.2 - 850 \times 0.3]$$
$$= 9.81[6256 - 200 - 255]$$
$$= 9.81 \times 5801 = 56907.81 \frac{N}{m^2}$$
$$= 56907.81\ Pa \cong \mathbf{56.9\ kPa}$$

3. $P_{atm} = \rho_m g h = 13600 \times 9.81 \times 0.750 = 100062\ N/m^2 = 100062\ Pa$

$P_{atm} = \mathbf{100.1\ kPa}$

4. The gauge pressure at state 1 is given by



$P_{atm} + \rho_l g h_1 = P_1$

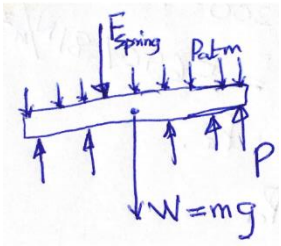
$P_1 - P_{atm} = \rho_l g h_1 = 28\ kPa$

$$\rho_l g h_1 = 28\ kPa; \quad \rho_l g = \frac{28\ kPa}{h_1}$$

The gauge pressure at state 2 is given by

$$P_2 - P_{atm} = \rho_l g h_2 = \frac{28\ kPa}{h_1} \times h_2 = 28\ kPa \times \frac{9\ m}{3m} = \mathbf{84\ kPa}$$

5. Balancing vertical forces, we write



$PA = P_{atm}A + W + F_{spring}$

$$\therefore P = P_{atm} + \frac{mg + F_{spring}}{A}$$

$$= 95\ kPa + \frac{4 \times 9.81 + 60}{35 \times 10^{-4} \times 10^3} = 95\ kPa + 28.35\ kPa \approx 123.4\ kPa$$

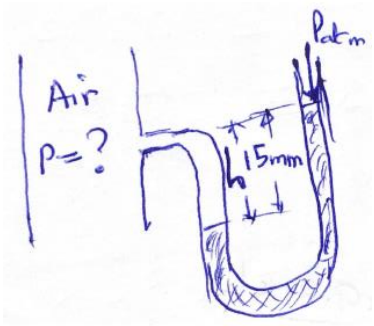
SELF ASSESSMENT 2-1 (Page 19)

1. $P = P_{atm} + \rho_m gh$

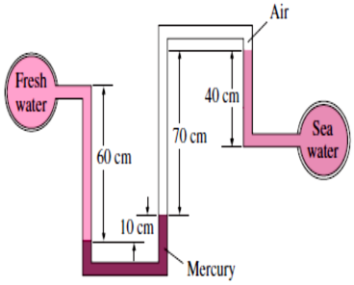
$= 100\ \text{kPa} + \dfrac{13600 \times 9.81 \times 0.015}{10^3}\ \text{kPa}$

$= 100\ \text{kPa} + 2.0012\ \text{kPa}$

$= \mathbf{102\ kPa}$



2. Starting with the pressure in the fresh waterpipe as point 1 and moving along the tube by adding as we go down or subtracting as we go up with the ρgh terms until we reach the sea water pipe point 2 gives



$P_{fw} + \rho_{fw} g h_{fw} - \rho_m g h_m - \rho_{air} g h_{air} + \rho_{sw} g h_{sw} = P_{sw}$

$P_{fw} - P_{sw} = \rho_m g h_m + \rho_{air} g h_{air} - \rho_{fw} g h_{fw} - \rho_{sw} g h_{sw}$

$= g[\rho_m h_m + \rho_{air} h_{air} - \rho_{fw} h_{fw} - \rho_{sw} h_{sw}]$

$= 9.81[13600 \times 0.10 + 1.15 \times 0.70 - 1000 \times 0.60 - 1035 \times 0.4]$

$= 9.81 - [1360 + 0.805 - 600 - 414]$

$= 9.81 \times 346.805 = 3402.16\ \text{Pa} = \mathbf{3.402\ kPa}$

3. $P_{A(abs)} - P_{B(abs)} = P_{gauge,A} = 140\ \text{kPa}$

$P_{B(abs)} = P_{absolute,A} - 140\ \text{KPa} = 267.7\ \text{kPa} - 140\ \text{kPa}$

$P_{B(abs)} = 127.7\ \text{kPa}$

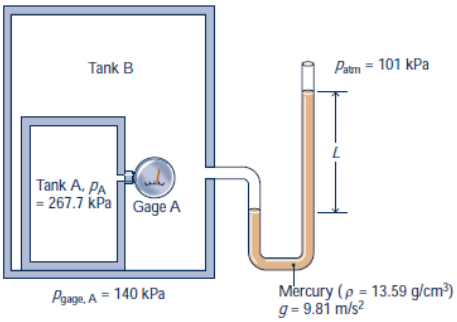
But $P_{B(abs)} = P_{atm} + \rho_m gL$

Hence, $127.7\ \text{kPa} = 101\ \text{kPa} + \dfrac{13600 \times 9.81 \times L}{1000}$

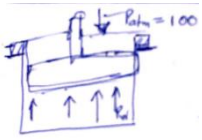
$\dfrac{26.7 \times 10^3}{13600 \times 9.81} = L$

$L = 0.2001\ \text{m}$

$L = \mathbf{20\ cm}$



TYPICAL UNIT 1 CALCULATION MCQ'S IN ME 265 - DETAILED SOLUTIONS



1. Balancing vertical forces, we write: $P_w A = Mg + P_{atm} A$

$$P_w = \frac{Mg}{A} + P_{atm} = \frac{100 \times 9.81}{0.01} + 100{,}000 = 198100 \text{ Pa} = \mathbf{198.1 \text{ kPa}}$$

2. $v = 1.5 \frac{m^3}{kg}$ is $> v_g$ at 200 °C $= 1.2721 \frac{m^3}{kg}$

hence its superheated vapour $1.08049 < 1.5 \text{ m}^3/\text{kg} < 2.1724$

$$\text{hence, } P_{required} = 100 + (200 - 100)\frac{(1.5 - 2.1724)}{(1.08049 - 2.1724)} = \mathbf{161.6 \text{ kPa}}$$

3. Energy balance for steam flow results in $q + h_{ground} = h_{top} + gZ_{top}$

$q = -100$ kJ/kg; $h_{top} = h_{g@125\text{ kPa}} = 2685.35$ kJ/kg

$$h_{ground} = 2685.35 + \frac{9.81 \times 100}{1000} - (-100) = 2796.33 \text{ kPa}$$

$$T_{ground} = T_{sat@2796.33\text{ kPa}} = \mathbf{163.54}\ ^\circ\text{C}$$

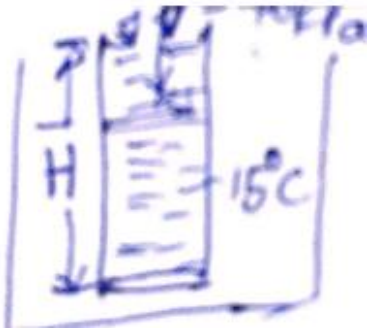
4. Initial specific volume $= v_1 = 0.5 \text{ m}^3/\text{kg}$

Interpolating for the pressure of the same specific volume of $0.5 \text{ m}^3/\text{kg}$

$$= 400 + \frac{(0.53422 - 0.5)}{(0.53422 - 0.42492)}(500 - 400) = \mathbf{431.31 \text{ kPa}}$$

5. $m = \rho \text{Volume} = \frac{\text{Volume}}{v} = \frac{AH}{v}$; $\quad v_{@15^\circ\text{C}} = 0.001001 \text{ m}^3/\text{kg}$

$$m = \frac{200 \times 10^{-4} \text{m}^2 \times 5\text{m}}{0.001001 \text{ m}^3/\text{kg}} = \mathbf{99.9 \text{ kg}}$$



6. $\frac{P_i V_i}{T_i} = mR = \frac{P_f V_f}{T_f}$

$$T_f = \frac{P_f V_f T_i}{P_i V_i} = \frac{350 \times 800 \times 10^{-3}}{150 \times 400 \times 10^{-3}} \times 300 \text{ K} = \mathbf{1400 \text{ K}}$$

7. $\Delta u = c_v \Delta T = 0.657 \times (1200 - 600) = \mathbf{394.2 \text{ kJ/kg}}$